Deon Quek

Applied AI & Backend Software Engineer

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PROFILE

Applied AI and backend engineer building governed agent platforms, knowledge systems and distributed services. Combines LLM tool execution, permission-aware retrieval and evaluation with reliable PostgreSQL-backed workflows, security boundaries and production-minded release engineering.

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL, JavaScript, Java, C++
Backend & APIs: Node.js, NestJS, Fastify, FastAPI, REST/OpenAPI
Frontend & Automation: React, Next.js, n8n, webhooks, OAuth
ML & Data Science: NumPy, pandas, scikit-learn, TensorFlow/Keras
AI & Agents: MCP, RAG, LLM APIs, tool calling, embeddings, evals
Data & Messaging: PostgreSQL/RLS, Redis, Supabase, queues, outbox
Testing & DevOps: Vitest, pytest, Playwright, Docker, GitHub Actions, Git
Architecture: Idempotency, event-driven design, approvals, audit

PROFESSIONAL EXPERIENCE

AI Automation & Backend Engineer — Splendify Jan 2025 – Present

- Designed document-ingestion and legal case-processing workflows with TypeScript, Python, PostgreSQL and n8n, integrating Gmail, Google Drive and Telegram for intake, parsing and audit review.
- Improved asynchronous job reliability through idempotent state changes, retry-safe processing and database constraints that reduce duplicate executions and make failures traceable.
- Containerized and migrated backend services to simplify deployment and reduce hosting overhead; built technical proofs of concept for enterprise customer discussions.

Founder & Software Engineer — Onyx Digital Jan 2024 – Apr 2026

- Delivered full-stack web applications, CRM integrations and lead-generation systems for SMEs using TypeScript, Next.js, React, Python and PostgreSQL.
- Led technical discovery, API design and deployment; automated manual business processes through webhook-driven integrations.

ENGINEERING PROJECTS

Mirai — Multi-Tenant AI Agents & Business OS Jan 2026 – Present

- Architected a multi-tenant AI business platform spanning persistent agents, a unified customer inbox, merchant commerce and the Atlas developer surface (MCP, REST/OpenAPI, SDK, CLI).
- Unified human and agent actions behind server-derived tenant identity, PostgreSQL row-level security, typed tools, policy checks, human approvals and auditable receipts.
- Engineered durable asynchronous execution with a transactional outbox, Redis Streams and workers, including retries and replay; verified ingress-to-orchestrator and inbox projection flows in a live local stack.
- Developed MiraiEval for invariant-based AI evaluation and evidence-backed release decisions (992 tests passing at programme integration); hardened staging, migrations and worker canary checks.

Verticals — Multi-Tenant Company Brain & Enterprise Search Feb 2026 – Present

- Extended Onyx Community Edition into a self-hostable Company Brain for connected business sources, permission-aware RAG, reranking, citations and source-provenance-backed retrieval.
- Built Python/FastAPI and Next.js capabilities around PostgreSQL tenant isolation, source/credential management, integration health checks and extensible MCP tools.

Heidi — Personal AI & Agent-Commerce Network Oct 2026 – Present

- Specified a Singapore-first agent-commerce network using A2A/MCP Shopfronts, versioned contracts, agent negotiation and human-approved commitments (pre-launch).
- Implemented early TypeScript/Fastify/PostgreSQL foundations with separate agent/merchant roles, row-level security, job queues, feature flags and spending controls.

EDUCATION

BSc (Hons) Computer Science — Machine Learning & Artificial Intelligence
University of London | 2026 | Upper Second Class Honours (2:1)

Diploma in User Experience & Design
Republic Polytechnic